TECHNICAL WHITEPAPER · NOVEMBER 2026

Real-time assistance on live healthcare calls

For ML engineers, AI engineers and MLOps practitioners

Streaming speech. Curated retrieval. Schema-driven extraction.

A streaming speech pipeline, a retrieval stack on a small embedding model, and a schema-driven encoder, plus how we measure whether any of it is right.

STATUS AS OF NOVEMBER 2026

OVERVIEW

Abstract: suggestions that a care manager decides on

The system listens to a two-party call between a care manager and a health-plan member, transcribes both sides in real time and puts suggestions on screen. It never makes a determination.

Use case A

Recommends documents from a curated library of about 136 items.

Use case B

Extracts structured facts from claim-denial calls.

Several results are negative and reported as such: a topic gate failed calibration, seven far-field audio front-ends failed, and a similarity threshold cannot separate right from wrong documents.

The method uses pre-registered decision rules, call-clustered statistics, counts on small families and a measured noise floor.

> Most open problems are corpus and labelling problems, not modelling ones.

DESIGN CONSTRAINTS

Five constraints shaped every design choice

Assistive only

Precision on screen matters more than recall. A wrong document costs attention on every call.

Nothing persisted

PHI in, no database, no transcript text on disk or in logs. Logs carry ids, counts and levels only.

No weights in the application

Speech, embedding and extraction run as separate processes behind narrow interfaces.

Latency target

Suggestions start within 5 s and settle within 3 s. Defining where the clock starts was the hard part.

Behaviour changes are flags

Default off, decided by operational and business review in user acceptance testing.

Design stance

Assistive only · Nothing persisted · Narrow interfaces · Measured latency · Default-off changes

RUNTIME

System architecture: every model runs out of process

The platform emits one media stream per call. An audio adapter runs one stereo speech session per call, orders finals by media time and sends turns to the orchestration service.

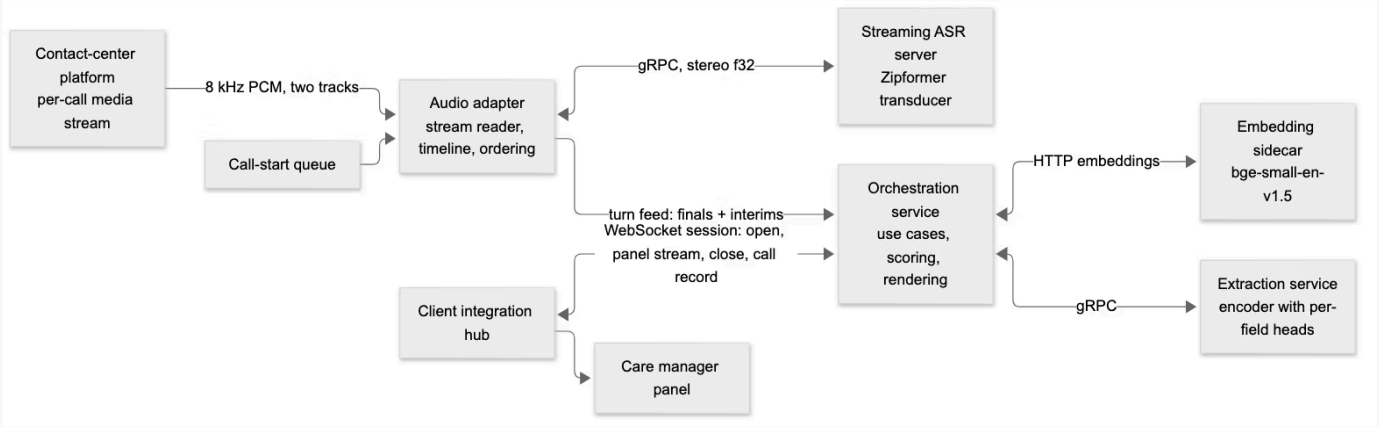


Figure 1. Runtime components.

The orchestration service calls the embedding sidecar or extraction service and streams view operations to the agent panel. At call end it sends a call record with no transcript.

Application-held artefacts

Schemas, lexicons, thresholds and the library index.

INTERFACES

Seven wire contracts, fingerprinted so they cannot drift

The contracts cover turn feed, contact media, transcript, member context, knowledge library, panel stream and call record.

1

Pydantic model

Generates each JSON Schema.

2

Canonical JSON

Strip descriptions, titles and examples.

3

SHA-256

Truncate to 16 hex characters.

4

CI test

Recompute every fingerprint.

Versioning follows meaning

Adding a field keeps the version. Removing one or changing meaning bumps it.

Field order is model input

Use-case schemas carry a separate prefix fingerprint that includes field order for the extraction encoder.

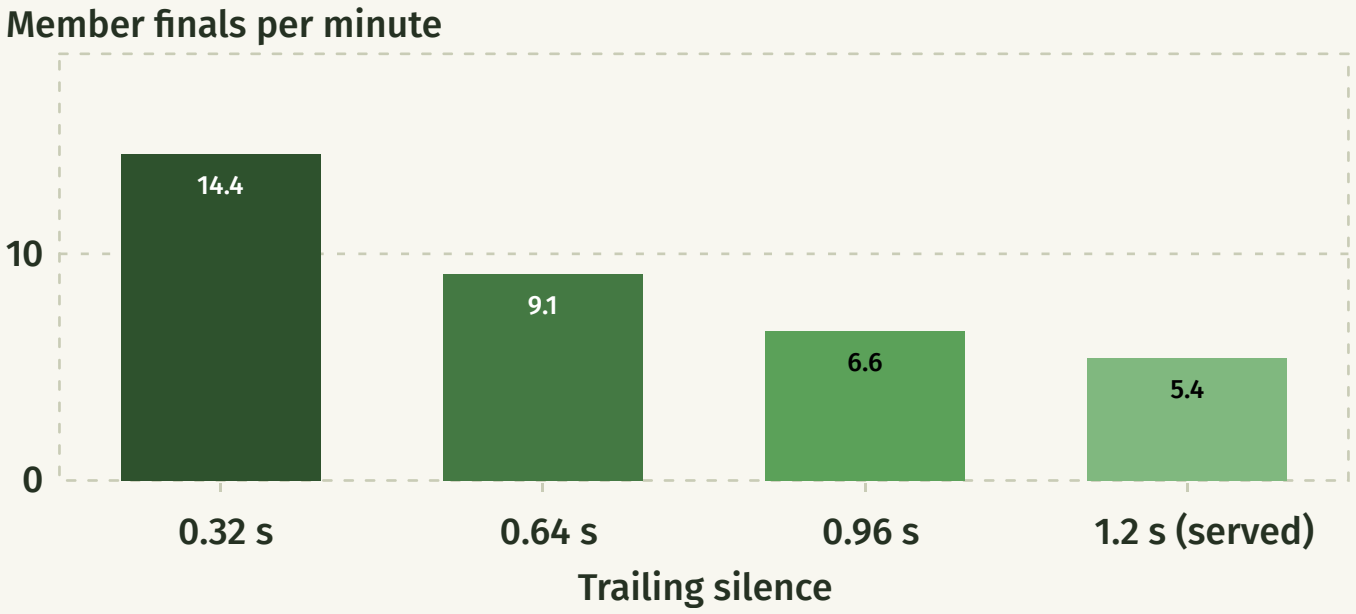
SPEECH

Speech layer: Zipformer wins on response time, endpointing is the lever

A streaming transducer service in Rust over ONNX serves the 2023-06-21 English Zipformer: about 340 MB, one model for every call.

Pipeline	Word error	p50	p95
Zipformer	0.269	32 ms	43 ms
Parakeet streaming	0.269	124 ms	140 ms

Member finals per minute by trailing-silence rule



Shorter pauses split turns into more, shorter segments. Production finalizes after 1.2 s trailing silence or 20 s continuous speech. The cap is a recommendation-quality parameter: it stays at 20 s until a human-judged comparison says otherwise.

AUDIO ADAPTER

Speaker attribution by channel, and ordering across two legs

Member and agent arrive as separate tracks in one stereo session per call. Speaker identity is never inferred.

Channel 0

Member

Channel 1

Agent

01

Validate the tracks

Labelled AAC, but carrying 8 kHz 16-bit mono PCM in 64 ms frames. Check rate, depth and channel count.

02

Set media start time

Use the first word's timestamp when word timings are present.

03

Reorder in a min-heap on media start time

Hold each final until the other leg advances past it, up to 2.0 s.

04

Emit ordered turns

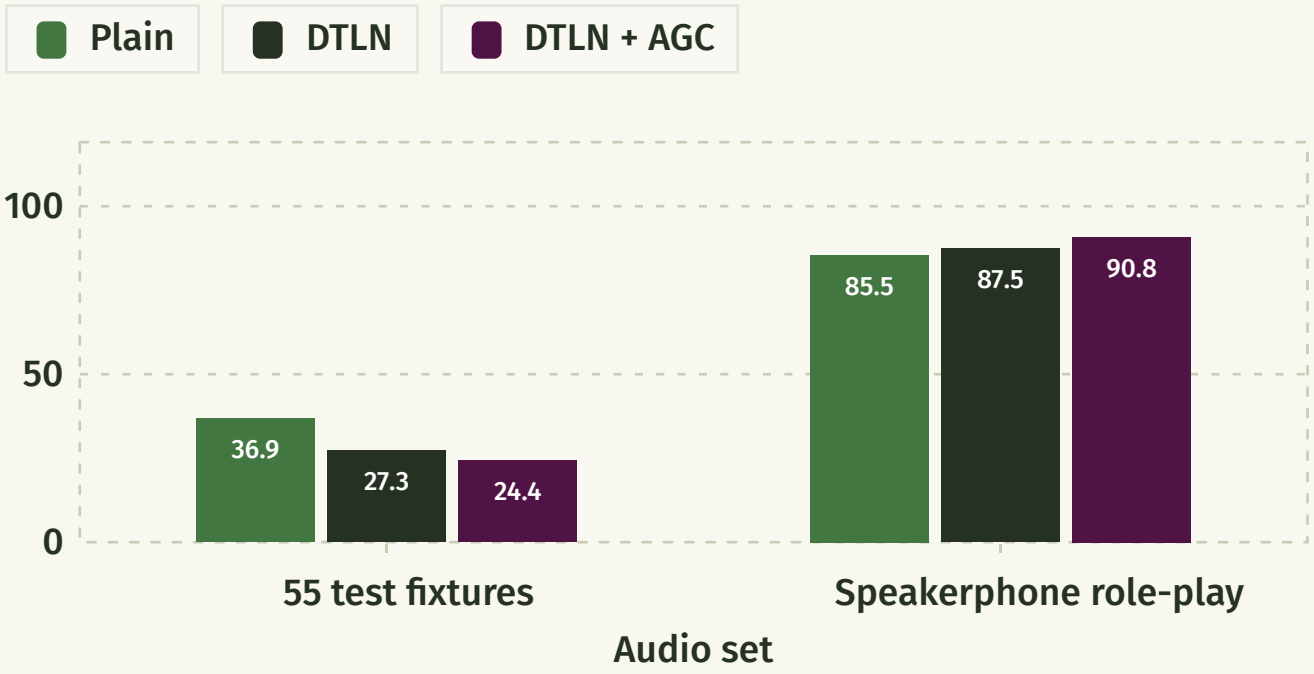
2 of 637 finals arrived late.

NEGATIVE RESULT

Far-field audio: a negative result reported in full

On our one real speakerphone recording, about 2 m and -34 dBFS, none of seven front-ends recovered the topic words: two AGCs, noise suppression with and without AGC, echo cancellation, GTCRN, DPDFNet, DeepFilterNet3, Silero VAD.

Word error rate (%)



55 test fixtures (mu-law, accented); speakerphone role-play, member channel.

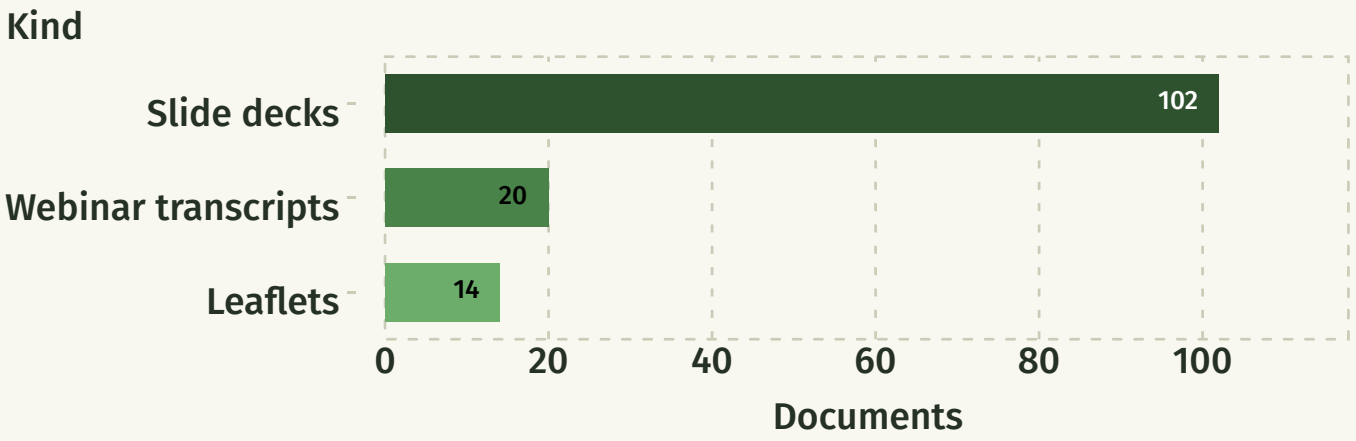
Echo cancellation worsened close-talk speech from 43% to 51% WER. Nemotron scored 58% versus Zipformer 91% on the speakerphone call at about seven times the CPU.

Phase 1 ships plain Zipformer, recommends headsets and logs audio levels as counts. Denoising remains a configuration switch pending tests on our own audio. The recogniser loses 30% of utterances under 0.5 s and 13% between 1 and 2 s.

USE CASE A

A curated library, embedded and indexed with guardrails

136 documents in 783 chunks. SMEs curated primary topic, trigger phrases, synonyms, priority and a do-not-surface flag: 758 trigger phrases, 185 single words, 1,545 phrases embedded with synonyms.



BAAI/bge-small-en-v1.5 provides 384 dimensions at about 67 MB, MIT, via fastembed in a sidecar with an OpenAI-compatible embeddings endpoint. The model is baked in; the hub is offline.

Embedding is symmetric today. Asymmetric prefixing, about 0.07 cosine difference, is likely better but requires every threshold to be re-fitted.

library_index/v1

The artefact contains chunk texts, unit vectors, embedder id, content hashes, topic map and a canary vector. A load check plus a 60 s canary must stay within cosine 1e-3. Pre-building cuts start-up from about 43 s to 2 or 3 s.

A canary failure drops the index and serves no list: a wrong list is worse than none.

RETRIEVAL PATH

From turns to spans: the in-call recommendation path

Fillers make up 32% of member turns, so a span is the unit of retrieval.

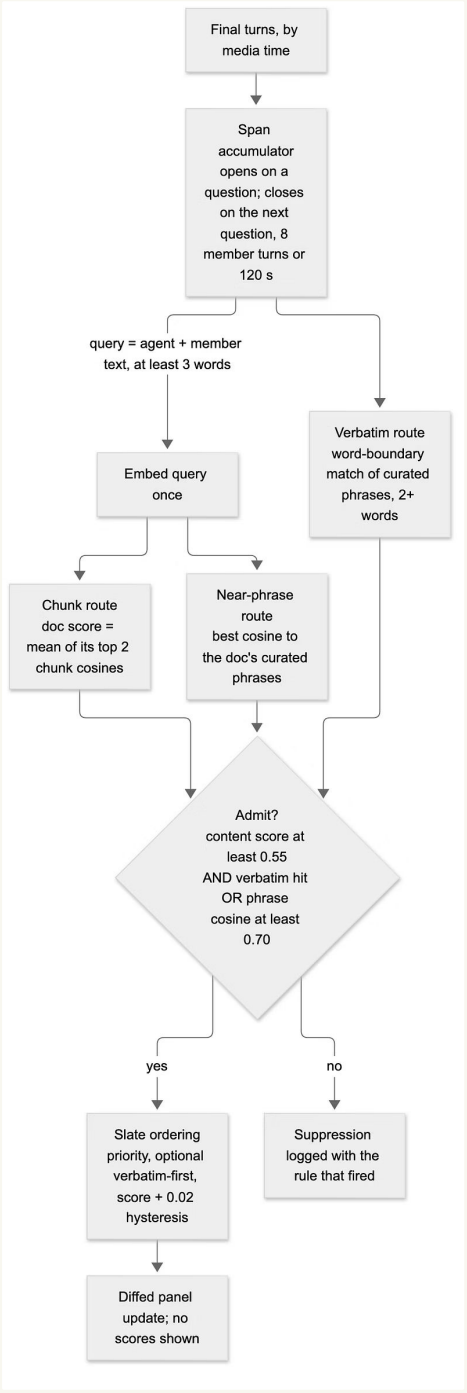


Figure 2. The in-call recommendation path. Scoring runs off the event loop in a worker thread.

Question-anchored spans

A span opens at a care manager question: a wh-word, auxiliary or modal after fillers, or a trailing question mark. It closes at the next question, after 8 member turns or after 120 s. Spans without member speech are dropped.

The query combines the span's agent and member text. Member-only admits almost nothing, since the care manager names the topic.

RUNTIME

One call, end to end: which component holds what

Contact-center media stream
One media stream per call

media stream

Audio adapter
One stereo speech session per call: channel 0 member, channel 1 agent. Min-heap reorder, up to 2.0 s hold.

finals, ordered by media time

ASR server
Zipformer, ONNX, Rust. About 340 MB, 24 calls per server.

turns

Orchestration service
Single replica. Holds schemas, lexicons, thresholds and the library index. Runs use cases. No persistence.

Extraction service
gRPC, per-field heads

Embedding sidecar
bge-small-en-v1.5, 384 dims, 2 s budget, circuit breaker

view operations

Client integration hub
Streams view operations to the agent panel

call end

End-of-call record
Sent by orchestration. No transcript.

Failure path: embedder down
Documents already shown stay on the agent panel, with a “not updating” notice.

Figure 1, redrawn as a single-call data flow. Solid arrows carry audio and decisions; the dashed box marks the embedder failure path.

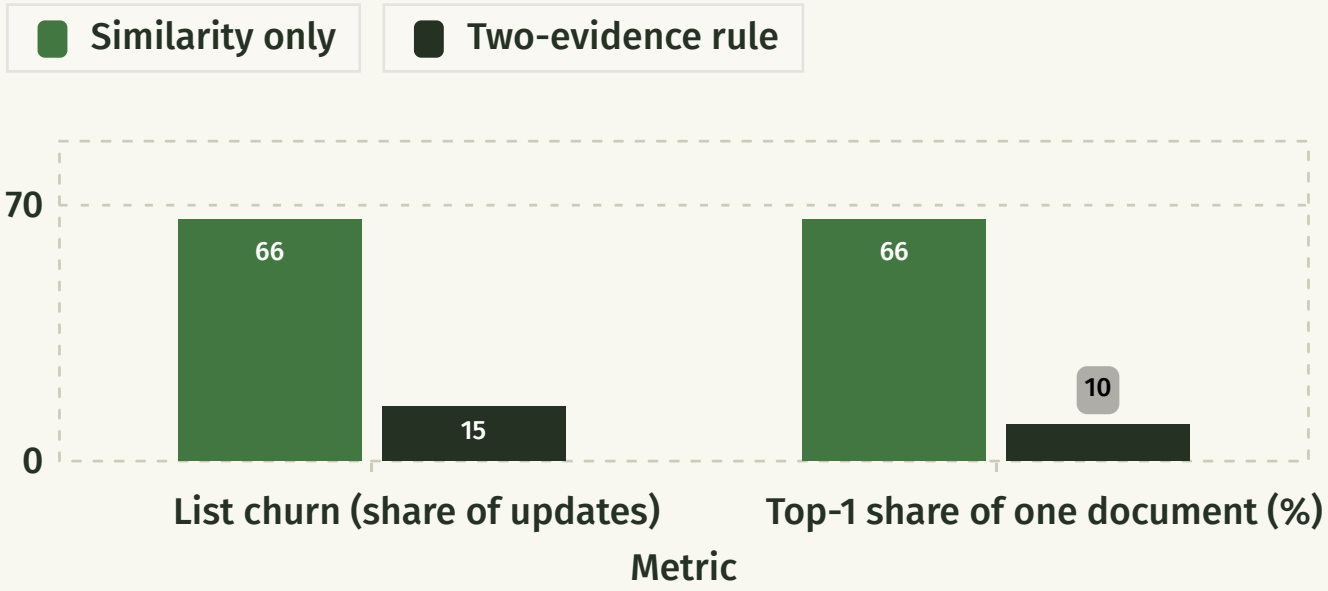
ADMISSION

Admission needs two kinds of evidence

In a 25-call, 543-span similarity-only sweep, one document took 66% of top-1 positions and the list churned on 66% of updates.

Content score ≥ 0.55

Plus either a verbatim curated phrase of two or more words, or phrase similarity ≥ 0.70.



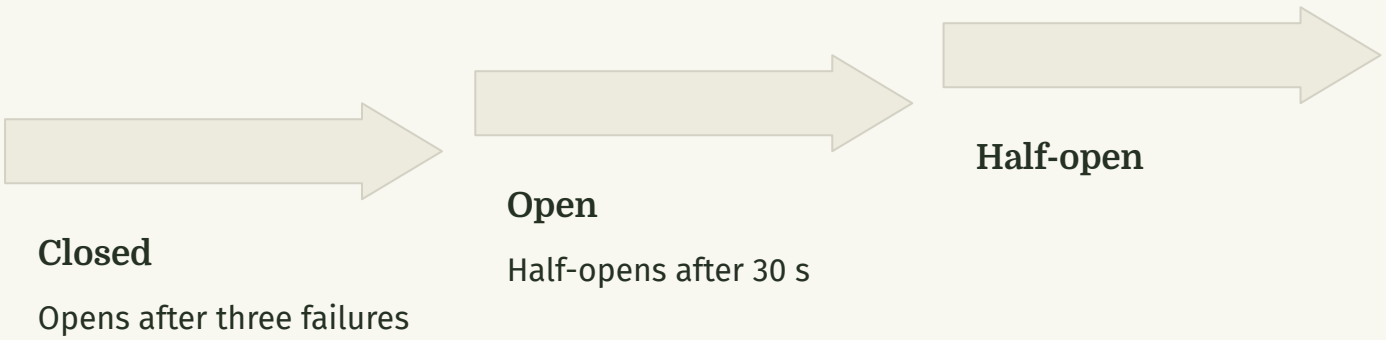
Churn fell from 0.66 to 0.15; top-1 share fell from 66% to 10%.

Supplemental documents need a verbatim hit; uncurated documents never show. 106 hand-maintained exact triggers across 17 documents admit verbatim only.

Ordering uses priority, optional verbatim-first, then score; 0.02 hysteresis; max-merge across spans. Documents are never withdrawn, and unchanged order produces no update.

Failure semantics and mechanisms deliberately not used

Embedding calls have a 2 s budget and no retries. Payload-too-large halves the batch.



An unreachable embedder never becomes an empty result. Shown documents stay with a “not updating” notice. There is no lexical fallback.

Mechanism	Status
Topic gate	k-NN vote, k=10. Leave-one-out accuracy 0.428 over 717 cases versus pre-registered 0.80 minimum. Action plans 0.32, campaign articles 0.39, webinars 0.38. Ships off.
LLM keep-or-drop judge	Built, not wired in. PHI requires an in-tenant model and larger latency budget.
Cross-encoder valence probe	Cancelled on a power calculation.

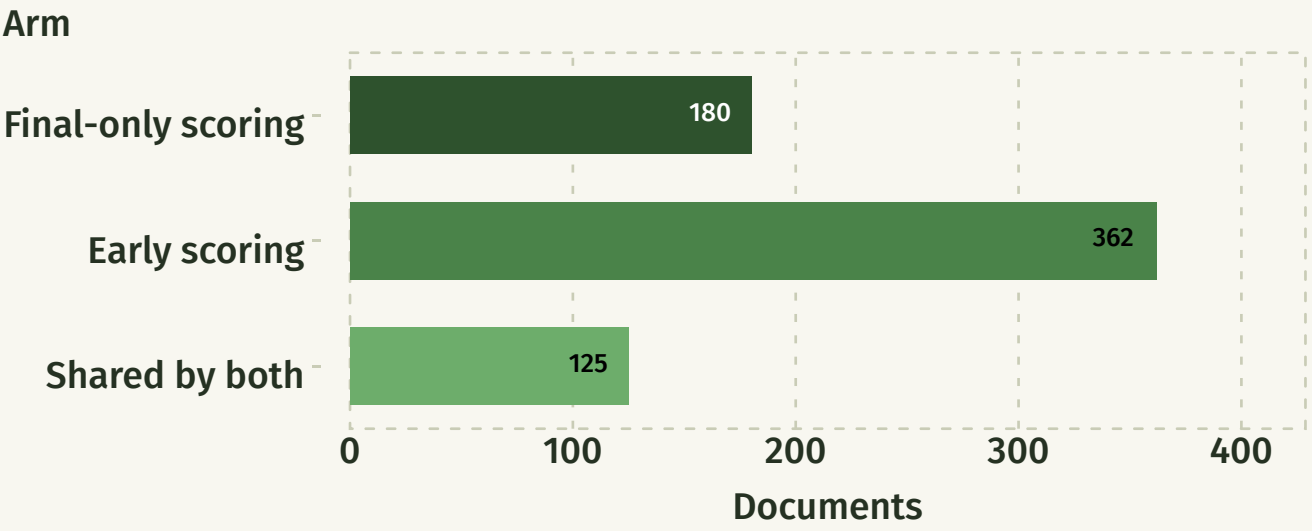
RECOMMENDATION EVIDENCE

What the recommendation data says

Every document admitted by a verbatim curated phrase was right. Every wrong one came through near-match, where both recall and errors live.

No threshold separates them: wrong best phrase similarity 0.743, right 0.736, correct caregiver documents 0.79 to 0.80.

Documents ever shown, 48 calls



Verbatim-first made the top five clean on all three live test calls, from 3, 1 and 1 wrong. One correct document dropped from fifth to sixth on two calls. Early scoring doubled documents ever shown and stays off.

Recorded queue announcements reached the scorer as care manager speech; this was fixed in call flow, not the model.

The ordering result rests on three role-played calls. The 48-call figures describe admission, churn and timing, not appropriateness.

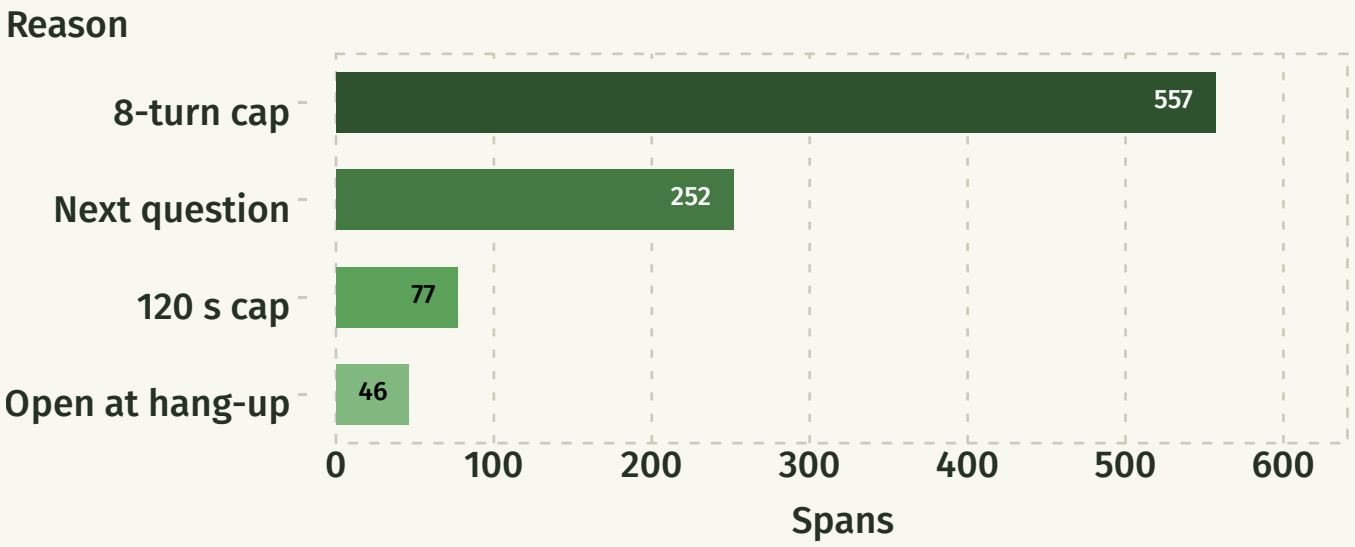
LATENCY

Latency anatomy: the system waits for the topic, not the model

Span scoring takes 20 to 50 ms: one query embedding plus a 3.7 ms dot product. Processing is about 0.5 s of the total; ASR finalize p50 is 0.13 s.

Clock	p50	p90
Member's last word to span close	4.8 s	20.2 s
Question start to panel update	6.3 s	14.3 s

Span closing reason (886 spans)



Only 22% closed within 3 s of the member's last word. The question detector fires on about a third of topic changes: topic-boundary detection is the real latency work.

Interim scoring needs two agreeing interim readings. On two role-played calls documents arrived a median 4.8 s and 13.4 s sooner, but doubles embedder traffic to about 46 req/s at 200 calls. It stays off until measured on real calls.

USE CASE B

Extraction served as a schema-driven encoder

A separate gRPC service is reached by a Rust client using PyO3, maturin and CPython 3.12. Extractors include a grammar-constrained generative model, ModernBERT QA, ModernBERT with per-field heads in production, and a hosted LLM. GLiNER2 retired in October.

01

Pydantic schema

Unknown keys forbidden. Field order sets token order.

02

Fields, descriptions and choices

Descriptions moved discriminator accuracy from 47% to 87%. Closed choices are one Literal; optionals get “not stated”; booleans pair with detail spans.

03

Per-field heads

With an abstain class. The checkpoint has 19 heads, trained when the schema had 20 fields; the schema now has 30.

Four per-workflow booleans fired together at up to 1.00, became one closed choice, then seven independent two-class flags.

EXTRACTION DECODING

Decoding by corroboration, and train-serve parity

Two views are retained: state, the last reading per source, and observations, every reading with score, provenance, scope and refusal.

One reading ≥ 0.9
Accept

→

Two readings ≥ 0.5
Accept by corroboration

Opening a workflow needs 0.95. Choice heads have a 0.5 floor. Rivals are judged by a 0.10 margin.

0.64 → 0.78
Precision with the rival margin

4 of 4
Workflows found

The largest defect was train-serve mismatch

The synthetic corpus uses 4,000 seeds, 52,569 samples and a fixed RNG seed; no real calls are seeds. Training had cased, contiguous identifiers, while production had lowercase normalized text. A lowercase payer read as silent where cased scored 0.992. The fix imports production's own normalizers.

Serving canonicalizes spoken digits and codes. Per-recogniser lexicons repair mishearings; 3 of 27 entries fire on the other recogniser.

EVALUATION

Evaluation methodology: pre-registered, call-clustered, noise-aware

Every user-visible mechanism gets a ledger decision rule before its run. Later rules are scored only on subsequently labelled calls. Calls are the unit: call-clustered paired bootstrap, detectable AUC about 0.80 at best, train/test split by care manager, small families as counts.

Rule	Mechanism	Outcome
R1	Harvest real phrasings into the lexicon	Failed, cancelled
R2	Cross-encoder valence probe	Cancelled, underpowered
R3-R5, R7, R8	Negation gate, dialogue-context judge, comparison endpoint, BM25 evidence, question-anchored topics	Registered, not run
R6, R6b, R6c	Slate topic gate, three designs	Failed calibration
R9	Span-level content slate with phrase agreement	Built, awaiting human judgement
R10	Verbatim-first ordering	Built behind a flag

Baseline semantic matcher, 328 labelled turns: 0.27 recall, 0.30 precision; fired on 36% managing fine versus 27% needing help.

Four seed-only extraction runs found 7 to 10 of 23 fields; gains under 3 fields are not reported. The workflow bar is 4 of 4 found, 0 of 3 negative controls, at most one question wrongly cleared.

Plumbing instruments: capture fingerprints, identity and differential oracles, golden replays, an expected-failure ledger (empty today), contract conformance suites.

MLOPS

MLOps: artefacts, builds and serving topology

01	02	03	04
PR checks	**Build**	**Image checks**	**Version-tag promotion**

Speech packages live in an internal feed; the embedder is baked into its image. The about 0.7 MB library snapshot and read-only index are tied by snapshot hash, embedder id and canary.

The pre-publish gate refuses invalid snapshots, libraries shrunk over 10%, duplicate document names, unresolved curated entries, mismatched index stamps and canary mismatches. Curation resolves by stable file name.

CI runs about 2,400 tests, three images and a vulnerability scan. Images are read-only, non-root, with capabilities dropped. Revision label and runtime build id must equal the commit; native wheels assert architecture and CPython 3.12 ABI.

Audio pods

Speech and adapter scale together. 960 s grace against a 900 s drain.

Two embedder replicas

Across zones.

One orchestration replica

Recreate; Guaranteed QoS. Production promotes a version tag, never rebuilds.

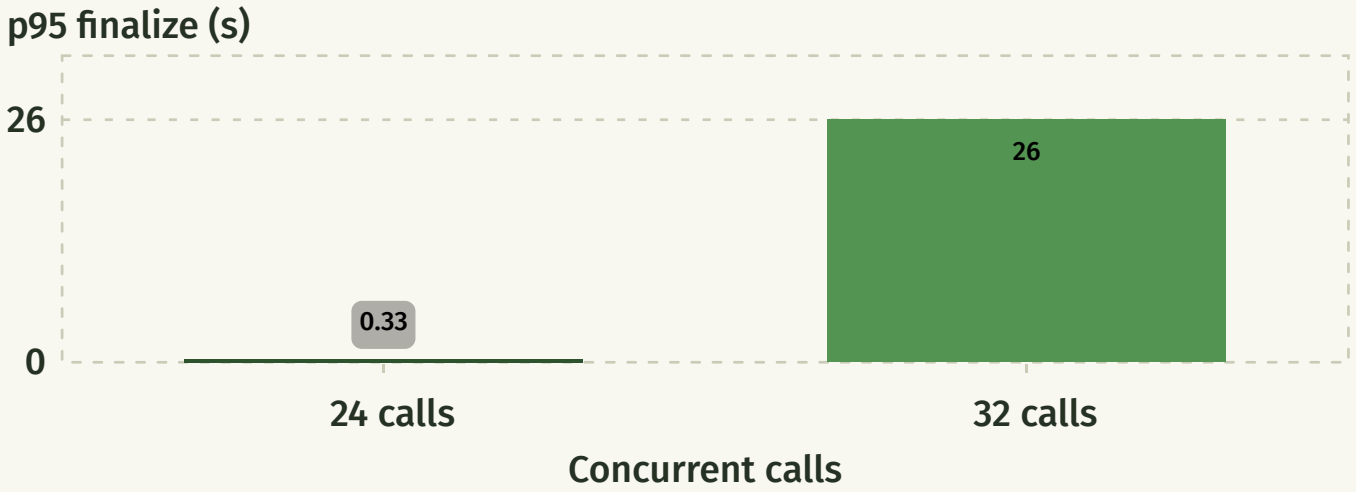
Logs carry ids, counts and levels. Start-up phrase count, 1,545 versus 1,046, confirms configuration took effect.

CAPACITY

Capacity and sizing

Typical design load is 50 to 100 concurrent calls, sized for 200. Measurements use an 8 vCPU VM with synthetic or role-played audio.

ASR finalize p95 (s)



Size ASR at 24 calls per server; at 32 calls, 26 s is growing. One intra-op thread moves the knee to 28 to 32. A 3-hour soak stays flat at about 2.6 GiB. With denoise, cap at 20 per pod.

The 2 vCPU embedder is clean to 36 req/s, p95 120 ms, with a knee by 48. At 200 calls it takes 2 to 13 req/s, about 46 with interim scoring; scale by replicas.

Orchestration uses about 8 MB per call. The adapter uses 39% of a core at 24 calls.

Production plan

Ten audio pods, two embedder replicas across zones, one service replica, about 92 vCPU. Ingress about 60 Mbit/s at 200 calls, provisioned 200.

LIMITATIONS · NEXT WORK

Limitations and where the work goes next

No human-judged precision on real calls yet. Near-match supplies both recall and errors; topic boundaries dominate latency; far-field 8 kHz speech is unsolved. Extraction corpus debt and symmetric embeddings remain. Role-played calls with one person in both roles are provisional.

1

Evidence

Expert verdicts on real-call lists against R9.

2

Latency

Measure interim scoring on real calls.

3

Latency

Topic-shift span closing, aiming at 3 s at p90.

4

Retrieval

Harden the library pipeline; build the index in the partner's weekly job.

5

Extraction

Regenerate and retrain the extraction corpus under the seed-variance protocol.

6

Retrieval

Register asymmetric prefixing, BM25 fusion and an in-tenant cross-encoder reranker first.

R9's real-call criteria

At least 60% appropriate; at least half of expert-expected documents; empty when nothing applies.

Figures come from replay and load-test tooling on synthetic, role-played or securely held recorded audio. No call content or personal information appears in this paper.